

D212: Data Mining II

Task 2: Dimensionality Reduction Methods

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Part I: Research Question

A1. Research Question

What are the key underlying dimensions or principal components that summarize patient characteristics and admission patterns, which can help inform more targeted and cost-effective healthcare strategies?

A2. Goal of the Data Analysis

The goal of this analysis will be to reduce the complexity of patient data by identifying the most influential variables that explain the majority of variability in patient characteristics and hospital admissions, in order to support strategic decisions on resource allocation, targeted care planning, and cost-effective service delivery.

Part II: Method Justification

B1. PCA Technique & Expected Outcomes

How PCA Analyzes Our Patient Dataset

Principal Component Analysis (PCA) is a statistical technique used to simplify complex datasets by reducing the number of variables (dimensions) while retaining most of the important information (variance) in the data (Singh, 2025).

In this analysis, Principal Component Analysis (PCA) will be applied as a dimensionality reduction technique to explore and summarize patterns within our patient dataset, to help us identify which combinations of variables explain the greatest differences among our patients and their hospital experiences.

The dataset contains multiple variables representing individual patient characteristics (e.g., age, income, vitamin D levels, full meals eaten, children, etc.) as well as healthcare utilization factors (e.g., doctor visits, hospital stay duration, total charges, and additional charges).

The first step involves standardizing all variables, to ensure that each contributes equally to the analysis regardless of its original scale. PCA will then be performed to transform the original set of potentially correlated variables into a smaller number of uncorrelated principal components that capture the most meaningful variation in the data.

The result of PCA will be a new set of dimensions, or principal components (PCs), which are linear combinations of the original variables. Each PC represents an underlying pattern or trend in the data. A scree plot will be used to determine the optimal number of components to retain using the elbow rule, which indicates which principal components capture the most meaningful variance in the dataset (Jaadi, 2024).

Expected Outcomes of the Analysis

The expected outcomes of this analysis are centered on uncovering the key underlying patterns within the patient dataset to better inform targeted and cost-effective healthcare strategies. By applying Principal Component Analysis (PCA), we aim to reduce the dimensionality of the data while retaining as much meaningful variation as possible. This allows us to simplify complex patient information into a smaller set of principal components that capture the most significant trends.

B2. Assumption of PCA

PCA assumes that the principal components capture the most meaningful structure in the data by maximizing variance (Whitfield, 2024). In other words, the most important information in the dataset is contained in the directions (components) where the data varies the most. If important patterns aren't associated with high variance, PCA might not capture those patterns effectively (Singh, 2024).

Part III: Data Preparation

C1. Variable Selection

These continuous (numeric) variables reflect health status, lifestyle behaviors, healthcare usage, and economic background - all key elements for understanding patient segments and drivers of hospital admission:

- Age (years)
- Income (USD)
- Children
- VitD_levels
- Doc_visits
- Initial_days (days)
- Full_meals_eaten
- vitD_supp
- TotalCharge (USD)
- Additional_charges (USD)

C2. Variable Standardization

Variable Standardization

To standardize the continuous data set variables identified in section C1, we'll use scikit-learn's StandardScaler. This will transform each feature (column) by removing the mean and scaling to unit variance.

```
# Initialize the scaler  
scaler = StandardScaler()
```

The scaler will be fitted to the 'df_subset' DataFrame to calculate the mean and standard deviation for each column (feature). These values will then be used to standardize each column by subtracting

the mean and dividing by the standard deviation. The resulting standardized values, where each column has an approximate mean of ~0 and standard deviation of ~1, will be stored in the 'scaled_data' NumPy array.

```
# Fit the scaler to the data and transform it
scaled_data = scaler.fit_transform(df_subset)
```

Since 'scaled_data' is just an array, we'll convert it back into a DataFrame so we can view and work with it a bit more easily.

```
# Convert scaled data to a DataFrame
df_scaled = pd.DataFrame(scaled_data, columns=df_subset.columns)
```

Finally, we'll print out that new DataFrame to verify that our variable standardization was successful.

```
# Check the means and standard deviations
print("Column Means:\n", df_scaled.mean())
print("\nColumn Standard Deviations:\n", df_scaled.std())
```

These printed values confirm that the dataset has been successfully standardized.

```
Column Means:
Age                1.147527e-16
Income             7.389644e-17
VitD_levels       -5.561773e-16
Doc_visits         3.161915e-17
Initial_days      -1.818989e-16
Full_meals_eaten  -7.815970e-17
TotalCharge        9.094947e-17
Additional_charges 9.094947e-17
Children           5.506706e-17
vitD_supp          3.694822e-17
dtype: float64
```

```
Column Standard Deviations:
Age                1.000005
Income             1.000005
VitD_levels       1.000005
Doc_visits         1.000005
Initial_days      1.000005
Full_meals_eaten  1.000005
TotalCharge        1.000005
Additional_charges 1.000005
Children           1.000005
vitD_supp          1.000005
dtype: float64
```

Cleaned Data Set

The cleaned dataset, 'd212_task2_transformed_medical_clean.csv', can be found within the main project directory and is also included as an attachment to this report.

Part IV: Analysis

D1. Matrix of ALL the Principal Components

After standardizing our data, we initialize a PCA model and apply the fit_transform function to the scaled data (df_scaled). This function identifies the directions of maximum variance and projects the original data onto these new directions, or principal components (Jaadi, 2024). The results are stored in a new matrix called 'pca_components', where each column represents a principal component, and each row represents a transformed version of the original data observation.

```
# Perform PCA
pca = PCA()
pca_components = pca.fit_transform(df_scaled)
```

The **principal component matrix** is then created by transforming the matrix of PCs into a DataFrame (df_pca) and displaying the PCA-transformed dataset. This new dataset expresses all observations in terms of principal components rather than the original variables.

```
# Create a DataFrame for PCA components
df_pca = pd.DataFrame(pca_components, columns=[f'PC{i+1}' for i in range(pca_components.shape[1])])
print(df_pca.head())
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	\
0	-1.394175	-0.781262	0.717999	1.344205	-0.825324	-0.095322	-0.386778	
1	-1.007013	0.535903	-0.085415	-0.221186	0.690230	0.559306	1.477619	
2	-2.160123	0.650647	-0.933919	-0.465136	0.761887	0.192652	-0.542437	
3	-2.409252	0.009420	-0.590140	-0.915494	0.240073	-1.451923	-0.046108	
4	-2.387166	-1.556694	0.262325	-1.910294	-0.341102	2.093849	0.304140	

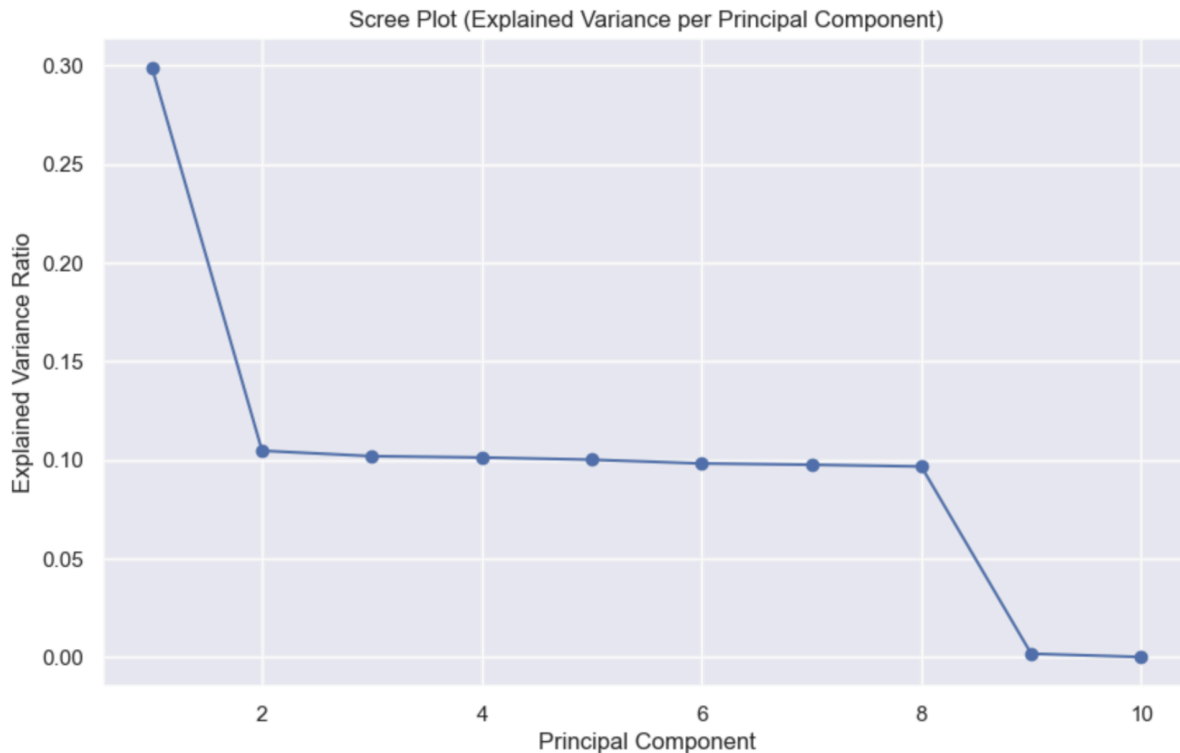
	PC8	PC9	PC10
0	-1.277356	0.153340	-3.522654e-17
1	0.094865	0.179924	1.093649e-16
2	-0.018224	-0.152329	6.071826e-15
3	-0.887192	-0.167808	2.215851e-16
4	1.212219	-0.162838	-4.707135e-17

D2. Total Number of Principal Components

Next, we want to identify the total number of principal components using the elbow rule to decide how many PCs are worth keeping. We extract the explained variance ratio for each PC from the fitted PCA object. Each value represents the proportion of the total variance in our data that's explained by the individual principal component (PC1, PC2, PC3...).

```
# Explained variance ratio = variance explained by each component
explained_variance = pca.explained_variance_ratio_
```

We then plot the variance for each PC, creating our scree plot. Following the elbow rule, we're looking for a point where the curve levels off (the "elbow") in our plot, which equates to how many PCs to keep (IBM, 2025).



Interpreting our scree plot, the “elbow” appears at Principal Component 2. Based on the elbow rule, this means we should keep 2 principal components. These two components (PC1 and PC2) explain the most substantial amount of variance and additional components offer only diminishing returns.

D3. Variance of Each Principal Component

The next step is to determine the variance explained by each of the two principal components selected in section D2. We’ll use a for loop to iterate twice, once for each component (PC1 and PC2), and print the explained variance ratio for each component. The explained variance ratio represents the actual percentage of variance explained by that specific component. Our result shows that **PC1 explains 29.86% of the total variance and PC2 explains 10.46% of the total variance.**

```
# Variance ratio (proportion of total variance explained)
explained_variance_ratio = pca.explained_variance_ratio_

# Variance of the retained components
num_components = 2

# Individual variance
for i in range(num_components):
    print(f"PC{i+1} explains {explained_variance_ratio[i] * 100:.2f}% of the total variance")
```

PC1 explains 29.86% of the total variance
PC2 explains 10.46% of the total variance

D4. Total Variance of Principal Components

To find the total variance of the principal components, we simply add up the variance from section D3 using the sum method. This shows that the first 2 principal components explain **40.32% of the total variance**.

```
# Cumulative variance
total_cumulative_variance = explained_variance_ratio[:num_components].sum()
print(f"Together, the first {num_components} principal components explain {total_cumulative_variance * 100:.2f}% of
```

Together, the first 2 principal components explain 40.32% of the total variance.

D5. Results of Data Analysis

Results

Circling back to my initial research question, this Principal Component Analysis (PCA) was conducted to identify the key underlying dimensions that summarize patient characteristics and admission patterns, with the goal of informing more targeted and cost-effective healthcare strategies. The analysis revealed that the first two principal components account for the most significant variation in the dataset, collectively explaining approximately 40.32% of the total variance.

- PC1, which alone explains 29.86% of the variance, appears to be primarily driven by admission-related variables such as initial days, total charges, and additional charges, suggesting that this component represents a “hospital resource utilization and cost” dimension.
- PC2, explaining an additional 10.46% of the variance, is more strongly influenced by factors such as age, nutritional intake (full meals eaten), and vitamin D levels, indicating a “patient health and nutritional status” dimension.

Together, these two components provide a meaningful summary of the data, capturing essential patterns in both healthcare utilization and individual patient profiles. These insights can help our hospital segment patient populations more effectively and tailor resource allocation strategies to reduce costs and improve care outcomes.

Part V: Attachments

E. Panopto Video Recording

Panopto video recording:

<https://wgu.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=6ac63c90-d573-4ee2-986c-b2b201225455>

F. Data & Code Sources

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